

Reverse Engineering a Four-Stroke Internal Combustion Engine

Section 2, Group 16

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Introduction

What is an engine?

An engine is a device that transfers the chemical potential energy stored in fuel into the kinetic energy of motion. It is found in cars, lawn mowers and even power tools like chainsaws.



Why we chose an engine



Engines are an integral part of our current civilization since they are responsible for getting us to and from most of our destinations. Despite this, most people don't understand how they work because they appear complex at the surface. In reality, the function of an engine can be broken down in a relatively comprehensive manner.

History of the internal combustion engine

- In 1876, Nikolaus Otto patented the four-stroke internal combustion engine which is the basis of most modern engines today.
- The reciprocating engine, which has one or more pistons that travel up and down, became the standard design for internal combustion engines, as opposed to the rotary engine.
- The one-cylinder engine provides relatively low power output, ideal for smaller engines like those in landscaping equipment.
- Over the years, combustion engines became more efficient with inventions like overhead camshafts, variable valve timing, and electronic fuel injectors.

Daimler and Maybach

- Gottlieb Daimler was a German inventor who developed high speed gasoline engines.
- His work focused on automobile engines, and later machines like lawn mowers.
- Wilhelm Maybach was also a German engineer.
- Daimler and Maybach worked together on the development of automotive engines.
- In late 1883, Daimler and Maybach patented the first of their engines fueled by Ligroin that achieved Daimler's goal of being small and running fast enough to be useful at 750 RPM.

Disassembly process

Engine removal

- After removing the oil and gasoline, the first step in our disassembly process was to remove the engine from the lawnmower chassis.
- The engine is held to the mower chassis by two 10mm and two 12mm bolts through the bottom
- After removing the bolts, the engine was easily lifted off the chassis.



First steps

- After the engine was freed, we first removed the fuel tank from around the engine. It was held on by four small screws.
- Next we removed accessory parts of the engine like the carburetor, muffler, pull-cord assembly, and ignition coil.
- After this, we removed bolts of various sizes from the exterior of the engine. There were 12 bolts on the crankcase, 8 on the cylinder head, and a variety of others on the exterior of the engine.
- The next step was to remove the crankcase cover as it prevented us from accessing the interior of the engine.



Engine disassembly - problem

- The flywheel and crankcase cover are held on by a 40mm nut to the engine's base.
- The nut was rusted to the crankshaft, and required more force than we could apply through a socket wrench in order to be removed.
- We dropped the engine, tried hammering at the bolt, but none of these methods were successful in removing the nut.



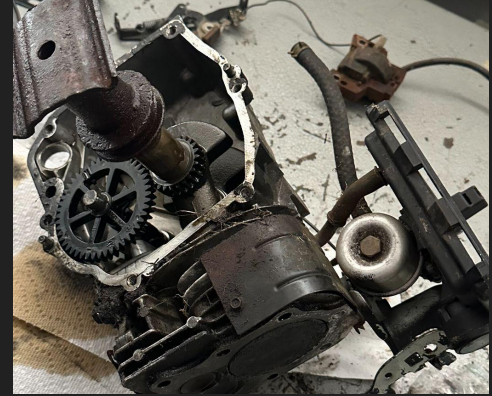
Engine disassembly - solution

- It wasn't long into the deconstruction process that we ran into this obstacle.
- After consulting Professor McLaughlin, we were instructed to meet with mechanical engineer Thomas Mealy, who then soaked the bolt in a rust-removing compound for three days.
- Upon receiving the engine back from Mr. Mealy, the bolt was easily removed, giving us access to the inside of the crankcase.



Final disassembly

- After removing the crankcase cover, we removed the governor/oil slinger and the camshaft
- All that remained at this point was the piston, attached to the connecting rod, attached to the crankshaft.
- The bolts holding the connecting rod to the crankshaft were stripped to the point where we couldn't remove them.
- Despite this, we had access to everything we needed in order to complete a reverse engineering analysis of the engine.



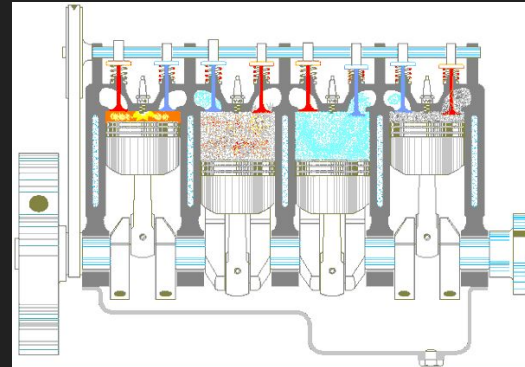
Engine details

Engine specifications - Briggs & Stratton 126T02-0220-B1

- Displacement: 190 cubic centimeters
- Bore: 52 millimeters
- Stroke displacement: 68.25 millimeters
- Oil capacity: 540-590 milliliters
- Operating speed: 2300-2800 RPMs
- Manufacture date: ~2005

Basic engine function

- This engine is a four-stroke internal combustion gasoline engine.
- Four distinct piston strokes: intake, compression, combustion, and exhaust.
- The piston will make two full strokes in one full cycle, and the crankshaft will make two full revolutions.
- The overall function of the engine is to convert the linear motion of the piston to rotational motion through the crankshaft.



Fuel requirements

It's important to adhere to the fuel octane requirements of any gasoline engine. If lower-than-recommended octane gasoline is used, the air-fuel mixture can ignite while the piston is still on its compression stroke. This is known as rod-knock or pre-ignition and can be damaging to the engine. This is the reason certain vehicles require higher octane “premium” gasoline. This engine requires a minimum of 87 octane unleaded gasoline.

Fuel system

The fuel tank (picture) is shaped to fit tightly around the engine itself to save space. There is a tube attached to the bottom of this tank on one end, and to the carburetor on the other. How the fuel is pulled from the fuel tank to the carburetor is explained on the [carburetor](#) slide.



Carburetor

The carburetor facilitates the mixing of air and fuel, ensuring the proper air-fuel ratio. Through the carburetor, the gasoline is vaporized because it ignites more easily in the form of vapor. A low pressure vacuum pulls atmospheric air through the air intake filter, and combines this air with gasoline that it pulls through a tube from the gasoline tank. The ratio of air to fuel depends on the engine and the operation altitude. The carburetor can be manually adjusted to manipulate the air-fuel ratio.



Air intake system

The air that the engine burns is atmospheric air pulled into the engine by a vacuum. The air is pulled through the air filter, sometimes called the air cleaner. The filter limits dust, dirt, and debris from entering the engine. It's important to change or clean the air filter as part of regular maintenance.



Lubrication requirements

Being a four-stroke engine, the engine requires oil to be filled separate from fuel. The manufacturer recommends 10W-30, or SAE 30 for higher temperatures, where the engine may consume more 10W-30. Oil levels should be checked regularly to ensure the engine isn't burning oil. The purpose of engine oil is to limit friction between the moving parts of the engine.

Oil dipstick tube

The oil dipstick tube is where the engine oil is poured into. The dipstick is attached the cap for the tube, and has a minimum and maximum oil level line to help gauge how much oil to add to the engine. For maximum accuracy, the dipstick level should only be read when the engine is cool and level.



Ignition coil

The ignition coil is made of two coils of copper wiring, which amplify the voltage from an electrical signal created by the spinning flywheel to activate the spark plug. As the electrical signal travels through the coils, they create a magnetic field which increases the current through the coil. By using Ohm's law, $\text{voltage} = \text{current} * \text{resistance}$, we can see that while the current increases, the voltage will also increase enough to allow for the spark plug to work.



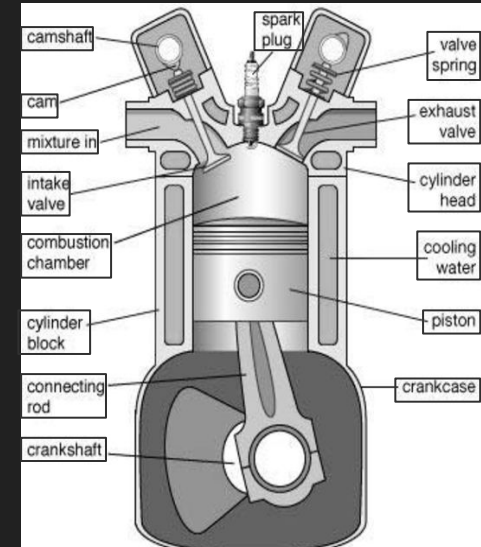
Spark plug

The spark plug ignites the fuel to begin the 'combustion' phase of the engine cycle. It does this by concentrating high voltages into a small point, which creates a miniature lightning strike to ignite the air-fuel mixture. Spark plugs generally run between 20,000 and 100,000 volts.



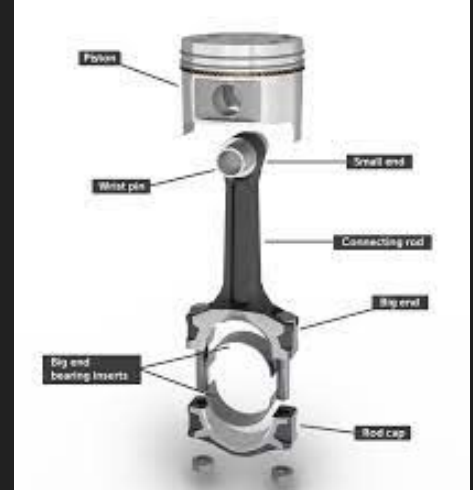
Cylinder

The cylinder of an internal combustion engine is where the potential chemical energy stored in the gasoline is converted to mechanical energy. The cylinder is the empty volume that the piston travels up and down it. When the intake and exhaust valves are closed, the cylinder is completely airtight to seal the combustion within the cylinder.



Piston

The piston's primary function is to convert energy from the combustion of fuel into mechanical energy. It does this by spinning the crankshaft via torsional force through the piston pin and connecting rod. The engine we're reverse engineering had one piston/cylinder, but larger engines can have as many as 16 in varying arrangements.



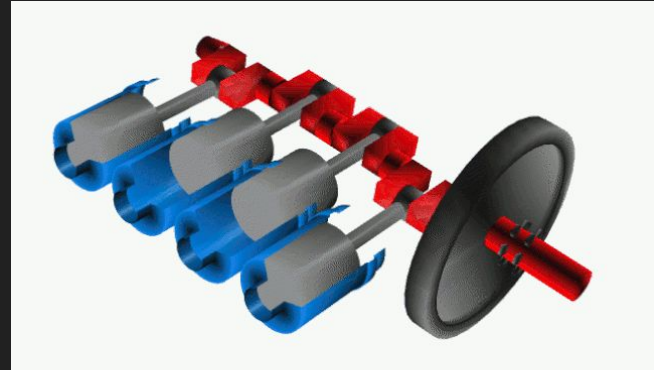
Cylinder head

The cylinder head is responsible for housing the intake and exhaust valves, and creating a seal at the top of the cylinder. These are generally made of cast iron to deal with the extreme temperatures they are subject to.



Crankshaft

The crankshaft is the point at which the piston's linear motion is converted to rotational motion, the two connected by [connecting rods](#). Being such a core component of the engine, the crankshaft is made out of very strong materials for high resistance to fatigue and wear. The semi-circle shaped protrusions on the shaft are counterweights to balance the engine. The crankshaft makes two full rotations for every cycle by the engine



Connecting rod

The connecting rod converts energy from linear to rotational. As the piston pumps up and down, the connecting rod transfers this energy into the rotation of the crankshaft. This component is put under a lot of stress in the form of tension, compression, bending, and buckling. To combat this, the connecting rod is often hollow and made of microalloyed steels for the added flexibility and strength.



Flywheel

The flywheel is a dense disk of metal attached to the end of the [crankshaft](#). The main purpose of the flywheel is to store rotational energy in the form of inertia. This energy is used to assist in the compression of the air-fuel mixture by the piston, which requires more energy than other strokes. It also aids in the expulsion of exhaust gases through the exhaust valve.



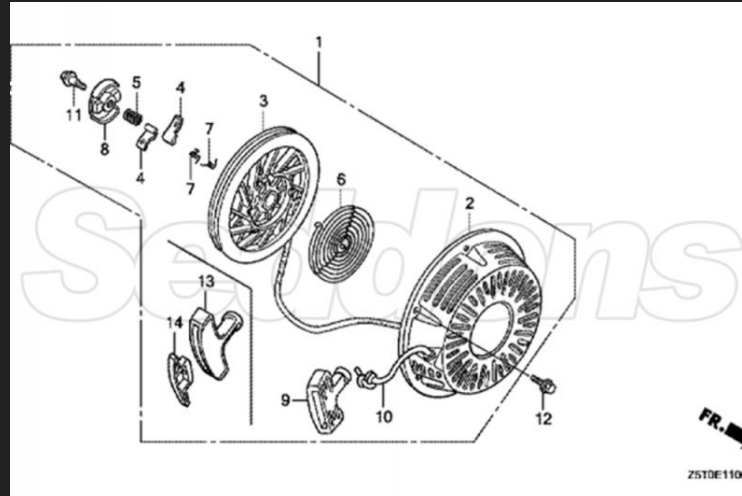
$RKE = \frac{1}{2} I * \omega^2$, where I is the moment of inertia and ω is angular velocity

ReadyStart® System

This engine features a temperature-controlled automatic choke, which requires no use of a primer or manual choke by the operator. The temperature sensor determines how much fuel the carburetor receives, to ensure easy starting in colder conditions. Air-fuel mixtures burn most efficiently when the fuel is vaporized, but when an engine is cold it's harder for gasoline to vaporize, so colder engines require more fuel per unit of air. To start the engine, all the operator has to do is rapidly pull the [pull-cord](#).

Pull-cord

The pull-cord is connected to the flywheel, so when the operator pulls the pull-cord, the whole engine is cycled. The pulling of the pull-cord also signals the ignition coil to send a charge to the spark plug, to ignite the air-fuel mixture and start the engine.



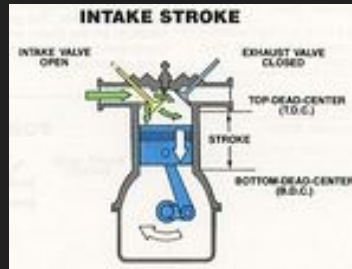
The intake valve

The intake valve is the valve by which the already mixed air-fuel mixture is pushed by atmospheric air pressure into the cylinder. [Lobes](#) on the rotating camshaft open the intake valve by pushing it down. Springs attached to the valve itself push it back closed when the cam lobes aren't pushing against it. Notably, the intake valve is larger than the exhaust valve to ensure as much clean air can enter the engine as possible after combustion.



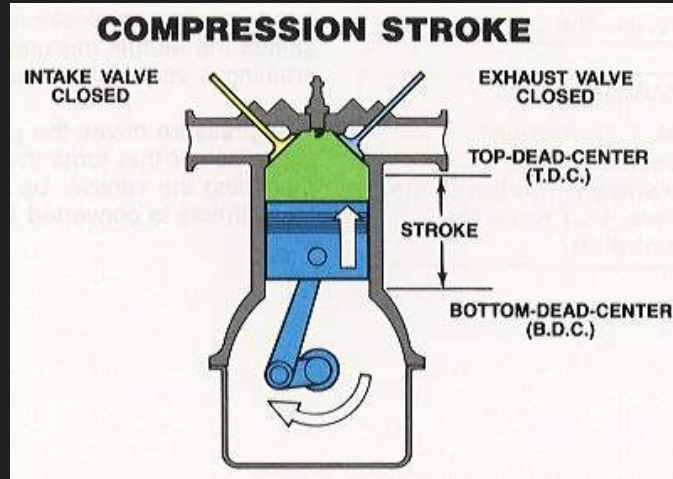
The intake stroke

On the intake stroke, the piston travels downward in the cylinder, and the air-fuel mixture fills the chamber via the intake valve, which is open on the intake stroke. The downward motion of the piston creates an area of low pressure within the cylinder, which pulls the air-fuel mixture through the intake valve. At the bottom of the intake stroke, the intake valve closes which seals the air-fuel mixture inside the cylinder.



The compression stroke

The compression stroke is the compression of the air-fuel mixture by the piston's upward motion. The rotational inertia of the [flywheel](#) helps maintain the piston's momentum on the compression stroke. On the compression stroke, the increase in pressure heats up the air-fuel mixture.



Compression

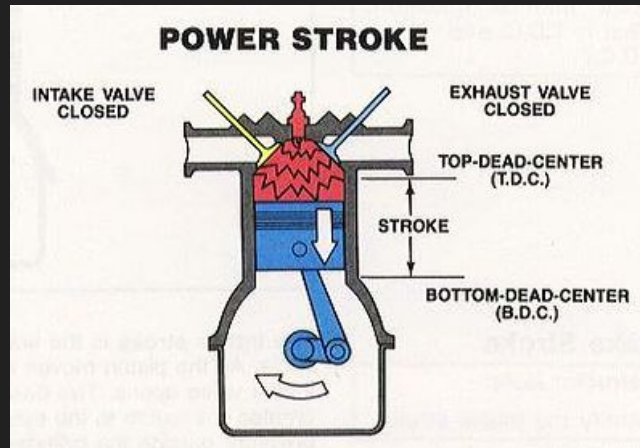
The higher the compression, the more energy released upon combustion, and the more fuel efficient the engine. Compression ratios vary from engine to engine, but engines with higher compression ratios are harder to start. The compression ratio is the ratio between the full cylinder volume with the cylinder bottomed-out (BDC), and the volume at the top of its path (TDC).

Ignition

Ignition occurs toward the end of the compression stroke, with the piston approaching TDC. The spark plug ignites the compressed air-fuel mixture as the piston reaches TDC.

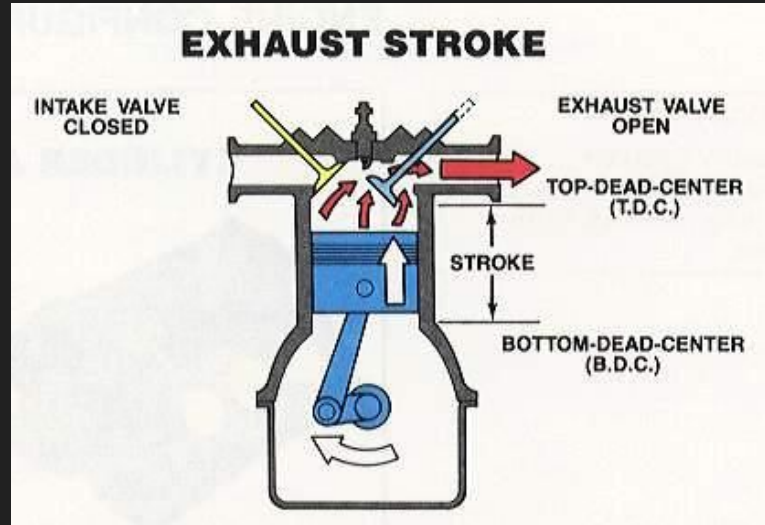
The combustion stroke a.k.a. the power stroke

The power stroke is initiated by the rapid expansion of gases inside the cylinder. These rapidly expanding gases force the piston head away from the cylinder head. Hence the name, the power stroke is where an engine's power comes from. The motion of the piston applies torque to the crankshaft via the connecting rod, causing it to rotate.



The exhaust stroke

The exhaust stroke is the final stroke in the engine's cycle. The returning upward momentum of the piston and the flywheel forces the exhaust gases out the open [exhaust valve](#). The end of the exhaust stroke marks one full cycle by the engine.



Exhaust valve

The exhaust valve exists to let the burned exhaust gasses out of the cylinder using the upward momentum of the piston. Lobes on the rotating camshaft open the exhaust valve by pushing it down. Springs attached to the exhaust valve push it back shut. The valve works by making and breaking a seal in the opening around the circular part of the valve.



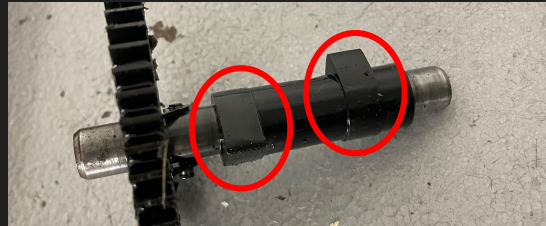
Camshaft

The function of the camshaft is to open and close the exhaust and intake valves. Some engines have one camshaft above the intake valves and another above the exhaust valves, others have a single overhead camshaft (SOHC). It's rotated by way of a geared connection with the crankshaft. The camshaft on a four-stroke engine rotates at one half the speed of the crankshaft, or one rotation per full engine cycle. This particular engine has one SOHC with lobes that open and close the valves.



Camshaft lobes

The camshaft lobes are fixed to the camshaft and push the flat circular heads of two pins, that push the valves open at the correct timing. The lobes sit at 0° and 270° on the camshaft. In terms of camshaft rotation, 0° to 90° is the intake stroke, so the intake valve must be open. 90° to 180° is the compression stroke and no valves should be open. From 0° to 180° , the crankshaft has made one full rotation, the piston a round-trip from TDC to BDC. 180° to 270° is the power stroke, where again no valves should be open. 270° marks the beginning of the exhaust stroke, and the exhaust valve is opened until the camshaft completes a full rotation at 360° .



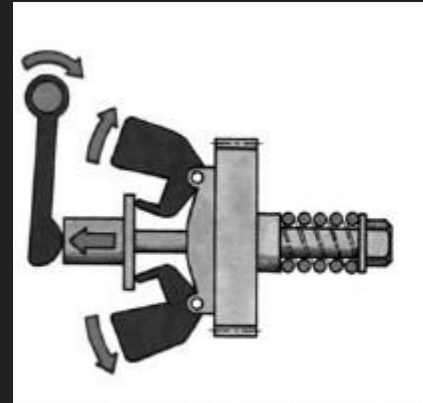
Oil slinger

The hole on the oil slinger sits on the end of the camshaft, with the gear teeth meshed perpendicularly against the camshaft gear teeth (meaning that the axes of rotation of the oil slinger and camshaft are perpendicular to each other). The oil slinger spins when the engine spins, and the teeth that point away from the axis of rotation splash oil around the inside of the engine, to lubricate effectively.



Centrifugal governor

The governor and oil slinger are one part that serves two functions. The governor is a cylinder shaped “spool” of metal with a flared base that rotates on the axis of the oil slinger. At rest, the two flyweights rest against the base of the governor, and it is held against the gear of the oil slinger by the governor cross shaft tab. As the oil slinger rotates at higher speeds, the flyweights open up and push the governor further away from the oil slinger, into the tab.



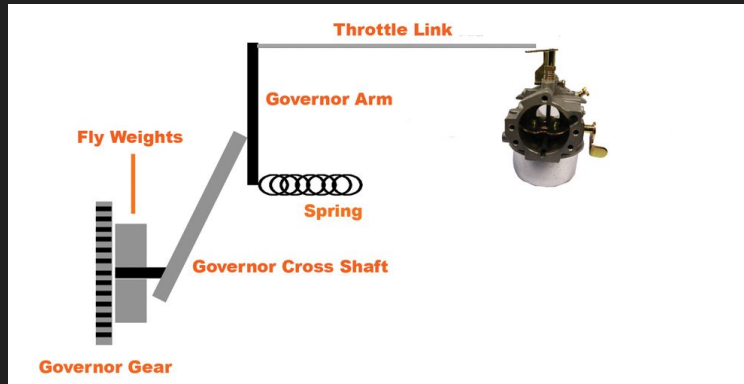
Function of the governor

At regular operating speeds, the governor is fully pressed outward against the governor shaft cross tab, and by [a series of mechanical connections](#), the throttle will be limited. When the engine experiences resistance and slows down, the governor will retract proportionally, and the throttle will open accordingly.



Governor mechanics

The governor cross shaft runs from the inside of the engine to the outside. At rest, a tab on this shaft is pressed against the governor spool by spring tension in the system (open throttle). This cross shaft is attached to an arm on the outside of the engine. This external arm is attached to the throttle control in the carburetor. When the tab is pushed against by the governor, meaning that the engine is reaching its operating speed, the arm rotates and reduces the throttle. When the engine speed slows down and the governor retracts, the spring tension will pull the whole system toward its resting state, increasing the throttle.



Muffler

The purpose of the muffler is to limit engine noise. The muffler works by using a series of baffles and/or tubes with perforations that complicate the path that the exhaust gases have to take out of the engine. Mufflers are designed with acoustics in mind, using opposing sound waves to cancel each other out - this is called destructive interference.



Exhaust

The exhaust from a gasoline engine consists of an array of chemical compounds.

- Nitrogen gas (71%)
- Carbon dioxide (18%)
- Water vapor (9%)
- Oxygen gas and other noble gases (~1%)

As well as toxic pollutants that make up less than one percent of exhaust

- Nitrogen oxide and nitrogen dioxide
- Carbon monoxide
- Unburned hydrocarbons.

Improving the design

Manufacturing of durable parts

Because of the high forces experienced by the crankshaft and connecting rod specifically, extensive research must be done to determine the most efficient materials and methods of manufacturing to use.

The crankshaft and connecting rod are made by either a casting or forging process. The forging process is more complicated and expensive, but forged materials have higher tensile strength than their cast counterparts.

Some crankshafts and connecting rods are made out of iron alloys, others steel alloys. The material of choice depends on the engines function. A small lawn mower engine like ours probably used cast iron, which is cheaper but also not as strong.



How do economic decisions impact manufacturing and design of combustion engines?

- **Materials:** Manufacturers seek materials that are durable and cost-effective. Materials like cast iron and aluminum are common, balancing performance and affordability.
- **Mass production:** The development of assembly line and mass production techniques by pioneers like Henry Ford greatly improved the affordability and availability of combustion engines. Economies of scale made engines more accessible to a broader consumer base as well.
- **Efficiency:** As fuel costs are a significant part of the total cost of vehicle ownership, manufacturers focus on designing engines that are more fuel-efficient. Economic considerations pushed them to create engines that provide better mileage, reducing the long-term operating costs for consumers.

How did economic decisions impact manufacturing and design of combustion engines? (continued)

- Environmental regulations: economic factors, including the cost of compliance with environmental regulations, influenced engine design. Stricter emissions standards led to the development of more complex engine technologies to reduce pollution while meeting economic constraints.
- Global market: Manufacturers face fierce competition in a global market. Economic factors influenced the design of engines that could perform well, reduce fuel consumption, and meet emission standards while remaining competitive in terms of pricing.
- Supply chain: Economic considerations also played a role in sourcing components and parts. Manufacturers often sought cost-effective suppliers and materials to maintain a competitive edge in the market.

Small improvements to the combustion engine

Hybridization: Combining internal combustion engines with electric powertrains in hybrid vehicles can significantly improve fuel efficiency and reduce emissions.

Advanced Fuel Injection: Utilizing advanced direct fuel injection systems and optimizing the timing can enhance combustion efficiency.

Turbocharging and Supercharging: Forced induction systems can boost engine power while maintaining fuel efficiency.

Variable Valve Timing: Implementing variable valve timing technology can optimize engine performance at different RPMs.

Lightweight Materials: Using lightweight materials like aluminum and carbon fiber can reduce the engine's weight and improve efficiency.

The future of engines

The future of the internal combustion engine is uncertain. The recent increase in popularity of electric lawncare equipment and cars has brought into question the environmental morality of gasoline engines because of how much emissions they produce.

Even though electric alternatives to gasoline engines are becoming more popular, the internal combustion engine will stay around for a long time because of how ingrained it is into our society; the majority of existing vehicles and tools are gasoline powered. Public confidence in electric cars and tools as a replacement for gas engine powered versions is still lacking, and in addition to this, much more research and development has been done on gasoline powered internal combustion engines.

Our advice to future REP groups...

For the duration of the project, we experienced few struggles that weren't our own fault. One namely mistake we made was not accounting for potential slow downs in the disassembly process. However, when we hit a roadblock in the deconstruction of our engine as we discussed earlier, it slowed us down considerably. Additionally, we didn't create a slideshow to document our work on until about two weeks before the projects due date. If we had started when the project was assigned, the workload would've been much more manageable. However, since we put it off, we had to spend our last few days working long hours every day. The biggest advice we can offer is to manage your time well. Never fall into the assumption that you have plenty of time before you have to worry about working on an assignment. Also, we would advise talking to other groups about their projects to figure out what works well and what doesn't.

End of Presentation Quiz

To reinforce the topics learned from this presentation, a questionnaire is included.

What is the main purpose of the Crankshaft?

- A. To convert reciprocating motion into rotary motion
- B. To regulate fuel and air intake
- C. To control exhaust emissions
- D. To provide steering control

Correct Answer is: A

A. To convert reciprocating motion into rotary motion

In a Gasoline Combustion Engine, Which Part of the Engine System is Responsible for Igniting the Air Fuel Mixture to Initiate the Combustion Process?

- A. Spark Plug
- B. Crankshaft
- C. Carburetor
- D. Camshaft

Correct Answer is: A

A. Spark Plug

What is the Purpose of the Exhaust Stroke?

- A. To draw in a fresh air-fuel mixture
- B. To compress the air-fuel mixture
- C. To expel burned gasses from the cylinder
- D. To generate power and drive the vehicle

Correct Answer: C

A. To expel burned gasses from the cylinder

Which Component in a Combustion Engine is Responsible for Connecting the Piston to the Crankshaft and Converting Reciprocating Motion into Rotary Motion?

- A. Camshaft
- B. Carburetor
- C. Throttle Body
- D. Connecting Rod

Correct Answer: D

D. Connecting Rod

Workload Split

Colin

- Disassembly
- Disassembly process slides
- Photos and Diagrams
- Component functionality
- Engine model research
- Speaker Notes

25%

Daniel

- Disassembly
- Organized help from Thomas Mealy
- Engine model research
- Component identification
- Component functionality
- Works cited

30%

Zack

- Disassembly
- Took Photos of project
- History of the engine
- Introduction
- Detailed the breakdown
- Quiz
- Speaker notes

25%

Henry

- Disassembly
- Works cited
- Quiz
- Pictures
- Component functionality

20%

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